Assignment -1

* Types of operating systems:- An operating system acts as an intermediary interface between computer hardware and the end-user. over decades, operating systems have evolved significantly to maximize cpu utilization, support multiple concurrent users and fit diverse hardware from factor.

1] Batch operating system:- Jobs with similar needs are batched and executed squentially

Usage:- Historically prominent in early mainframe systems. currently utilized for batch automation pipelines requiring no human intervation, such as automated credit card biling, payroll processing, and historical data archiving.

Advantages:- 1] High processing throughput:- Reduces total overhead because the system loads and executes similar jobs continously without switching contexts mid-task.

2] Unattended Automation:- Large blocks of computa-tions can be scheduled to run overnight or during off-peak hours without manual human oversight.

Disadvantages:- 1] zero Direct Interaction:- The user has no capability to input parameters or interact with the application during runtime.

2] complex Debugging:- If an error occurs in a specific job, it fails completely and diggostic logs are only made available after the entire batch finishes processing.

2] Multiprogramming operating system:- Multiple jobs reside in memory; os switches when one waits for I/O.

Usage :- Serves as the foundation for modern enterprise servers and multi-user computing environments looking to maximize computing throughput.

Advantages :- 1] High CPU Efficiency :- Minimizes expensive processor ~~p~~ idle time by ensuring a backup job is always ready for computation whenever an I/O block occurs.
2] Optimized Turnaround :- Significantly lowers the total turnaround time required to execute a cluser of mixed-workload programs.

Disadvantages :- 1] Complex memory management :- Demands robust allocation policies and fragmentation control to successfully accommodate multiple active process simultaneously in RAM.
2] Scheduling overhead :- Requires specialized CPU scheduling algorithms to decide prioritization when shifting execution control between memory tasks.

3] Time-Sharing Operating System :- An extension of multipro-gramming where the processor is allocated among multiple interactive users via rapid scheduling.

Usage :- Widely applied in massive mainframe computers. Central enterprise computing clusters, and cloud environments serving hundreds of concurrent user terminals.

Advantages :- 1] Rapid Respone Time :- provides instantaneous interactive feedback to users, typically responding to command inputs in fractions of a second.
2] Resources Equality :- prevents any single

user or dominant process from hogging system resources indefinitely, ensuring uniform service delivery.

Disadvantage :- 1] High security Risk :- Merging multiple concurrent data sessions introduces risk of cross-process memory leaks, demanding absolute process isolation.

2] Single point Failure :- If a catastrophic core Kernel crash occurs, all connected users lose their active sessions instantly.

4] Multitasking operating system :-

Uses :- 1] personal computers
2] Laptops
3] Smart phones

Advantages :-
1] Saves time
2] Better productivity
3] Efficient cpu usage

Disadvantages :-
1] Requires more memory
2] May slow down with many programs
3] Complex management

5] Multiprocessing operating system :-

Use :-
- Scientific computing
- Engineering application
- Data centers.

Advantages :-
- High speed
- Better performance
- Increased reliability

Disadvantages :-
- Expensive hardware
- complex design
- Difficult maintenance

6] Real-Time operating system :-

uses :-
- medical equipment
- Air traffic control
- Robotics

Advantages :-
- Fast response
- High reliable
- Accurate performance

Disadvantages :-
- Expensive
- complex programming
- Limited multitasking

7] Distributed operating system :-

uses :-
- Cloud computing
- Scientific research

- Large organizations

Advantages :-
- Resource sharing
- High performance
- Reliable

Disadvantages :-
- complex design
- Expensive setup
- Network dependency

8 Network operating system :-

Uses :-
- File sharing
- printer sharing
- office networks

Advantages :-
- centralized management
- Easy resource sharing
- Better security

Disadvantage :-
- server failure affects the network
- costly
- Requires skilled administrator.

9] Multiuser operating system:

Use :-

- Many people can use one computer system
- users can work at the same time.
- Files and printers can be shared.

Advantages :-

- Efficient use of hardware resources
- Supports collaboration among multiple users
- Better security and centralized management.

Disadvanatges :-

- more complex to manage.
- Requires powerful hardware
- A system failure can affect all connected users.

10] Mobile operating system :-

Uses:-

- Runs mobile apps.
- Connects to the internet
- play music and videos

Advantages :-

- Easy to use.
- supports many apps.
- Connects to the internet.

Disadvantages :-

- Limited battery life

- Small screen size
- Limited storage compared to computers.

* Types of system calls:-

1] Process control system calls

- These system calls are used to create, execute, terminate, and manage processes.

2] File Management system calls:

- These system calls are used to create open, read, write, close and delete files.

3] Device Management

- control and communicate with hardware devices.

4] memory Management

- Allocate, free and manage memory.

5] Information Management

- These system calls used to obtain or modify system and process information.

6] Communication system calls

- These system calls ~~control~~ allow communication between processes.

7] protection and security system calls

- These system calls control user permissions and protect system resources.

* Definationes of system calls

1] fork()

→ create a new child process by copying the parent process

2] vfork()

→ create a child process that shares the parent's memory until it executes a new program or exits

3] clone()

→ create a child process that shares the parent's memory until it executes a new program or exit

4] execve()

→ Replace the current process with a new program.

5] exit()

→ Terminates a process after performing cleanup.

6] _exit()

→ immediately terminates a process without cleanup

7] wait()

→ waits until any child process finishes execution

8] waitpid()

→ waits for a specific child process to finish.

9] waitid()

→ waits for process state changes and returns detailed information

10] kill()

→ Sends a signal to a process

11] getpid()

→ Returns the process ID (PID) of the current process

12] getppid()

→ Returns the parent process ID (PPID)

13] nice()

→ changes the scheduling periority of a process.

14] setpriority()

→ set the periority of process

15] getpriority()

→ Return's the priority of process.

16] open()

→ opens an existing file.

17] creat()

→ create a new file.

18] close()

→ close a new file.

19] read()

→ Read data form file

20] write()

→ writes data to a file.

21] pread()

→ Reads data from a specific file.

22] pwrite()

→ writes data to a specific file position without changings the file pointer

23] lseek()

→ Changes the current position of file pointer

24] Stat()

→ Return information About a file.

25] fstat()

→ Returns information about an open file.

26] lstat()

→ Return information about a symbolic link

27] access()

→ check the access permission of a file.

28] unlink()

→ Deletes a file from the file system

29] rename()

→ changes the name of a file or directory

30] mkdir()

→ creates a new directory

31] rmdir()

→ Removes an empty directory

32] dup()

→ creates a duplicate of a file descriptor

33] dup2()

→ duplicates a file descriptor to a specified descriptor

34] fcntl()

→ performs operations on file descriptors, such as changing flages or locking.

35] fsync()

→ writes all modified file data & meta-data to the disk

36] fdatasync()

→ writes modified data to the disk but minimzes metadata updates

37 truncate()

→ changes the size of a file using its file name.

38 ftruncate()

→ changes the size of a file using its file descriptor.